Chapter 10. Estimating Population Size and Nonresponse

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1 Capture–Recapture Estimation of N

To estimate the number N of fish in a lake, catch n_1 fish, mark them and release them; later catch a second sample of n_2 fish and count the number m of marked fish among them.

Definition 10.1: Assumptions

1. The population is closed (no births, deaths or migration between the samples).
2. Each sample is an SRS from the population (marking does not change catchability).
3. The two samples are independent.

The proportion of marked fish in the second sample should resemble that in the population:

$$\frac{m}{n_2} \approx \frac{n_1}{N} \quad \Longrightarrow \quad \hat{N} = \frac{n_1 n_2}{m}$$

(the Lincoln–Petersen estimator), where n_1 is the size of the first (capture) sample, n_2 that of the second (recapture) sample, and m the number of marked elements in the second sample.

1.1 Capture–recapture as a ratio estimator

Theorem 10.2

$\hat{N}$ is a ratio estimator of a population total, and

$$\widehat{\mathrm{Var}}(\hat{N}) = \frac{n_1^2 n_2^2 (n_2 - m)}{m^3 (n_2 - 1)} \approx \frac{n_1^2 n_2 (n_2 - m)}{m^3}.$$

Derivation

Define for every element $y_i = 1$, so $\tau_y = \sum_{i=1}^{N} y_i = N$, and $x_i = 1$ if element i is marked, 0 otherwise, so $\tau_x = \sum_{i=1}^{N} x_i = n_1$ is known. The recapture sample of size n_2 gives

$$r = \frac{\sum_{i=1}^{n_2} y_i}{\sum_{i=1}^{n_2} x_i} = \frac{n_2}{m}, \qquad \hat{N} = \hat{\tau}_y = r\tau_x = \frac{n_2}{m} n_1.$$

From Ch. 5, $\widehat{\mathrm{Var}}(\hat{N}) = \tau_x^2 \widehat{\mathrm{Var}}(r) = n_1^2 \widehat{\mathrm{Var}}(r)$ with

$$\widehat{\mathrm{Var}}(r) = \frac{1}{\bar{x}^2} \cdot \frac{N - n_2}{N} \cdot \frac{1}{n_2} \cdot \frac{1}{n_2 - 1} \sum_{i=1}^{n_2} (y_i - r x_i)^2,$$

and we take the fpc ≈ 1. Now

$$\bar{x} = \frac{m}{n_2}, \qquad \sum_{i=1}^{n_2} (y_i - r x_i)^2 = \sum y_i^2 - 2r \sum x_i y_i + r^2 \sum x_i^2 = n_2 - 2\frac{n_2}{m} m + \frac{n_2^2}{m^2} m = \frac{n_2(n_2 - m)}{m},$$

using $y_i^2 = y_i = 1$ and $x_i^2 = x_i$. Therefore

$$\widehat{\mathrm{Var}}(r) = \frac{n_2^2}{m^2} \cdot \frac{1}{n_2} \cdot \frac{1}{n_2 - 1} \cdot \frac{n_2(n_2 - m)}{m} = \frac{n_2^2 (n_2 - m)}{m^3 (n_2 - 1)},$$

and multiplying by n_1^2 gives the result.

1.2 Maximum likelihood

Definition 10.3

Let $f(\mathbf{x} \mid \theta)$ be the joint pdf of the sample $\mathbf{X} = (X_1, \ldots, X_n)$. Given $\mathbf{X} = \mathbf{x}$, the function $L(\theta \mid \mathbf{x}) = f(\mathbf{x} \mid \theta)$ of θ is the **likelihood function**. If $L(\theta_1 \mid \mathbf{x}) \geq L(\theta_2 \mid \mathbf{x})$, then θ_1 is more plausible than θ_2 in light of the data. The **maximum likelihood estimator** is $\hat{\theta}(\mathbf{x}) = \arg\max_{\theta \in \Theta} L(\theta \mid \mathbf{x})$.

Example 10.4: Normal mean

For $X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\theta, 1)$,

$$\ell(\theta) = \log L(\theta \mid \mathbf{x}) = -\frac{n}{2} \log 2\pi - \frac{1}{2} \sum_{i=1}^{n} (x_i - \theta)^2, \qquad \ell'(\hat{\theta}) = \sum (x_i - \hat{\theta}) = 0 \;\Rightarrow\; \hat{\theta} = \bar{x}.$$

Theorem 10.5: Large-sample properties of the MLE

Under regularity conditions,

1. (consistency) $\lim_{n \to \infty} P(|\hat{\theta}_n - \theta| \geq \varepsilon) = 0$ for every $\varepsilon > 0$;
2. (asymptotic normality) $\sqrt{n}(\hat{\theta}_n - \theta) \to N\big(0, 1/I(\theta)\big)$, i.e. $\hat{\theta}_n \approx N\big(\theta, \frac{1}{nI(\theta)}\big)$, where $I(\theta) = \mathrm{E}\big[\big(\frac{\partial}{\partial \theta} \log f(X \mid \theta)\big)^2\big]$ is the Fisher information of one observation.

Theorem 10.6: MLE of N

Model the recapture sample as $X_i = 1$ (marked) or 0, $X_i \overset{\text{iid}}{\sim} \mathrm{Ber}(p)$ with $p = n_1/N$, $i = 1, \ldots, n_2$. Then

$$\hat{N}_{\mathrm{MLE}} = \frac{n_1 n_2}{m}, \qquad \mathrm{Var}(\hat{N}_{\mathrm{MLE}}) \approx \frac{N^2 (N - n_1)}{n_1 n_2}, \qquad \widehat{\mathrm{Var}}(\hat{N}_{\mathrm{MLE}}) = \frac{n_1^2 n_2 (n_2 - m)}{m^3}.$$

Derivation

With $\sum_{i=1}^{n_2} X_i = m$,

$$L(N) = \prod_{i=1}^{n_2} \Big(\frac{n_1}{N}\Big)^{X_i}\Big(1-\frac{n_1}{N}\Big)^{1-X_i} = \Big(\frac{n_1}{N}\Big)^{m}\Big(1-\frac{n_1}{N}\Big)^{n_2-m},$$

$$\ell(N) = m\log n_1 - m\log N + (n_2-m)\log(N-n_1) - (n_2-m)\log N, \qquad \ell'(N) = \frac{n_2-m}{N-n_1} - \frac{n_2}{N}.$$

Setting $\ell'(N)=0$: $N(n_2-m) = n_2(N-n_1)$, so $\hat N = n_1 n_2/m$.

Fisher information. For one observation,

$$\frac{\partial}{\partial N}\log f(x\mid N) = \frac{1-x}{N-n_1} - \frac{1}{N} = \frac{n_1/N - x}{N-n_1} = \frac{p-x}{N-n_1},$$

$$I(N) = \mathrm{E}\Big(\frac{p-X}{N-n_1}\Big)^2 = \frac{p(1-p)}{(N-n_1)^2} = \frac{n_1}{N^2(N-n_1)}.$$

Hence $\mathrm{Var}(\hat N_{\mathrm{MLE}}) \approx \big[n_2 I(N)\big]^{-1} = \frac{N^2(N-n_1)}{n_1 n_2}$. Plugging in $\hat N = n_1 n_2/m$:

$$\frac{(n_1n_2/m)^2\,(n_1n_2/m - n_1)}{n_1n_2} = \frac{n_1^2n_2^2}{m^2}\cdot\frac{n_1(n_2-m)}{m}\cdot\frac{1}{n_1n_2} = \frac{n_1^2 n_2(n_2-m)}{m^3}.$$

The ratio-estimator and likelihood approaches agree: $\widehat{\mathrm{Var}}(\hat\tau_y) \approx \widehat{\mathrm{Var}}(\hat N_{\mathrm{MLE}})$.

Example 10.7: Illustration

If $n_1 = 200$ fish are marked and a second catch of $n_2 = 100$ contains $m = 20$ marked fish, then $\hat N = 200\cdot 100/20 = 1000$ and

$$\widehat{\mathrm{Var}}(\hat N) = \frac{200^2\cdot 100\cdot 80}{20^3} = 40{,}000, \qquad 2\sqrt{40000} = 400,$$

so N is estimated as 1000 ± 400.

2 Nonresponse

Nonresponse (missing data) biases the estimation of population parameters. Divide the population into a response stratum R and a nonresponse stratum:

	size	mean
Respondents	N_1	μ_1
Nonrespondents	N_2	μ_2
	$N = N_1 + N_2$	μ

Theorem 10.8: Nonresponse bias

If $\bar y_1$ is unbiased for the respondents' mean μ_1, then as an estimator of μ,

$$\mathrm{E}(\bar y_1) - \mu = \frac{N_2}{N}(\mu_1 - \mu_2).$$

Proof. $\mu = \frac{N_1}{N}\mu_1 + \frac{N_2}{N}\mu_2 = \frac{N_1+N_2}{N}\mu_1 - \frac{N_2}{N}\mu_1 + \frac{N_2}{N}\mu_2 = \mu_1 - \frac{N_2}{N}(\mu_1-\mu_2)$. ∎

The bias is small only if the nonresponse rate N_2/N is small or respondents and nonrespondents are alike ($\mu_1 \approx \mu_2$). Increasing the sample size does not reduce it.

2.1 A conservative interval for a proportion

Suppose n_1 respondents are an SRS from the N_1 respondents of the population, so a 95% interval for their proportion p_1 is

$$\hat{p}_1 \pm 2\sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1 - 1} \cdot \frac{N_1 - n_1}{N_1}}.$$

Nothing is known about the nonrespondents' proportion p_2, except $0 \le p_2 \le 1$. With $w_1 = N_1/N$ and $w_2 = N_2/N$, $p = w_1 p_1 + w_2 p_2$. Take the worst cases $p_2 = 0$ for the lower limit and $p_2 = 1$ for the upper limit:

$$\hat{p}_L = w_1\Big(\hat{p}_1 - 2\sqrt{\tfrac{\hat{p}_1\hat{q}_1}{n_1-1}\tfrac{N_1-n_1}{N_1}}\Big) + w_2 \cdot 0,$$
$$\hat{p}_U = w_1\Big(\hat{p}_1 + 2\sqrt{\tfrac{\hat{p}_1\hat{q}_1}{n_1-1}\tfrac{N_1-n_1}{N_1}}\Big) + w_2 \cdot 1.$$

Whenever the interval for p_1 covers p_1, $[\hat{p}_L, \hat{p}_U]$ covers p whatever p_2 is, so $P(\hat{p}_L \le p \le \hat{p}_U) \ge 0.95$. The width grows by w_2: a high nonresponse rate makes the interval uninformative, which is why reducing nonresponse (call-backs, follow-ups) matters more than increasing n.